

NAVLE Review

Ruminant Study Guide

A complete review of cattle, sheep, goat
and camelid medicine.

Forestomach & Metabolic · Enteric & Calf Diarrhea · Respiratory & BRDC

Neurology · Reproduction & Abortion · Mastitis & Udder

Musculoskeletal & Foot · Dermatology · Vector-Borne & Systemic

Parasitology · Small Ruminant Medicine · Camelids

Toxicology · Production Medicine & Regulatory

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HOW TO USE THIS GUIDE

Yellow boxes are board pearls and buzzword-to-diagnosis shortcuts. **Blue boxes** are look-alike discriminators. **Red boxes** flag genuine clinical dangers.

R = reportable **FAD** = foreign animal disease **Z** = zoonotic

1 | Forestomach, Abomasum & Metabolic Disease

1.1 Functional Anatomy

Structure	Detail
Passage of food	RETICULUM (honeycomb) → RUMEN (left side) → OMASUM → ABOMASUM (the true, glandular stomach).
Normal rumen pH	6 (roughly 6.0–6.8). It drops toward 5 in grain overload.
Rumenotomy	Best location: the LEFT PARALUMBAR FOSSA. Two indications: hardware disease (traumatic reticuloperitonitis) and grain overload / toxic plant removal .
Paravertebral block	Anesthetizes T13, L1, and L2 . Needle placement is over the transverse processes of L1, L2, and L4 for the proximal technique.
Dehorning	Block the CORNUAL NERVE — a branch of the zygomaticotemporal branch of the trigeminal (CN V). Sinusitis follows if you open the frontal sinus diverticulum when dehorning an adult.

1.2 Forestomach & Abomasal Disease

Condition	Features	Management
Rumen acidosis / grain overload	High-energy grain diet → <i>Streptococcus bovis</i> and <i>Lactobacillus</i> proliferate → lactic acid accumulates → rumen pH falls from 6 to 5 , killing the normal microbial population.	Alkalinizing agents, fluids, rumen transfaunation, rumenotomy if severe. Sequelae: rumenitis → liver abscesses , laminitis.
Rumenitis and liver abscesses	<i>Fusobacterium necrophorum</i> enters the portal circulation through damaged rumen wall. The classic feedlot sequela of grain overload.	Tylosin in feed is the standard PREVENTIVE ; an established abscess responds poorly to any antimicrobial. Note that <i>F. necrophorum</i> is also the agent of foot rot and calf diphtheria — one bug, three diseases.
Caudal vena cava thrombosis	The full chain: high-producing cow → grain overload/ rumenitis → <i>F. necrophorum</i> into portal circulation → liver abscess → erodes into the caudal vena cava → septic thrombus showers the lungs → pulmonary abscess RUPTURES INTO AN AIRWAY.	Three signs: right-sided heart failure · EPISTAXIS · HEMOPTYSIS. Grave prognosis; often found dead from exsanguination.
Vagal indigestion	Outflow failure at the pylorus (or reticulo-omasal orifice) from vagal nerve damage — often post-hardware disease or an abscess.	Two results: indigestion (papple-shaped abdomen) and BRADYCARDIA . The bradycardia is the tell.
Abomasal impaction	Two findings: HIGH CHLORIDE in the rumen (abomasal chloride refluxes forward) and "internal vomiting." Produces a hypochloremic, hypokalemic metabolic ALKALOSIS with paradoxical aciduria.	Fluids with chloride and potassium; surgery for true impaction.
Abomasal ulcer	Bleeding ulcers produce melena and profound anemia.	Best treatment: BLOOD TRANSFUSION — the cow bleeds out into her own GI tract. Perforating ulcers cause peritonitis.
Cecal dilation / volvulus	Predisposed during the FRESH period (early lactation), when the cow is on a high-concentrate diet with negative energy balance.	Right flank typhlotomy or medical management.

Condition	Features	Management
Hemorrhagic bowel syndrome	Located in the JEJUNUM . Associated with <i>Clostridium perfringens</i> type A and <i>Aspergillus</i> . Intraluminal blood clot causes obstruction.	Surgery; grave prognosis.
Right displaced abomasum (RDA)	An RDA LACKS two signs that an LDA has — the left-sided ping and the classic left-side location. RDA pings on the right and progresses to volvulus.	Surgical correction; RDA/volvulus is an emergency.
Atresia coli	The HOLSTEIN is at increased risk. Calf fails to pass meconium; abdominal distension.	Poor prognosis; often associated with rectal palpation of the amniotic vesicle early in gestation.
Rectal prolapse — four grades	1. Mucosa alone prolapsed 2. All layers of the rectum prolapsed 3. Intussusception of the descending colon into the rectum 4. Intussusception of the small colon into the rectum	Grades 1–2 are reducible; 3–4 require surgery. In cattle the two most common are grades 1 and 2.

1.3 Metabolic Disease

Condition	Pathophysiology & Signs	Treatment
Grass tetany / grass staggers	HYPOMAGNESEMIA in a lactating cow on lush pasture . Lush spring pasture is high in nitrogen and potassium, which inhibit magnesium absorption . Five signs: restlessness · staggers · over-alertness · excitability · falls down and dies . Muscle twitching, nervousness.	IV magnesium (with calcium); then oral supplementation and pasture management. Note the second cause of "staggers": <i>Phalaris</i>/canary grasses — stiff gait, hyperexcitability, nystagmus, head tremors, falling and flailing.
Fat cow syndrome / subclinical fatty liver	Over-conditioned cow in negative energy balance mobilizes fat into the liver.	Oral propylene glycol or IV dextrose — provide glucose precursors.
Post-parturient hemoglobinuria	A sign of PHOSPHORUS deficiency . Intravascular hemolysis with red urine in a fresh cow.	Phosphorus supplementation; transfusion if severe.
Hypovitaminosis A	Seen in WINTER or DROUGHT — no green forage. Three consequences: reproductive failure with low conception rates · vision loss / night blindness · immune dysfunction with persistent diarrhea .	Vitamin A supplementation.
Copper deficiency	Caused by MOLYBDENUM excess (molybdenum binds copper). Signs: " copper coat " — faded, rusty hair — with SPECTACLES around the eyes — plus diarrhea . In young animals: broken bones despite adequate vitamin D .	Copper supplementation; correct dietary molybdenum and sulfur.
White muscle disease	Selenium and/or vitamin E deficiency . Affects skeletal AND cardiac muscle .	Chemistry: elevated CK and AST. Selenium is a cofactor for GLUTATHIONE PEROXIDASE — measuring GSH-Px in whole blood is a subtle way to check selenium status. Treat with injectable selenium/vitamin E.
Zinc deficiency	Contributes to rain rot / dermatophilosis . In goats causes parakeratosis and poor coat.	Zinc supplementation.

Condition	Pathophysiology & Signs	Treatment
High mountain (brisket) disease	Pulmonary hypertension at altitude → right heart failure and brisket edema.	Screen with a PAP score: a good score is < 41 mmHg; above 46 mmHg the animal is susceptible. Treatment is to MOVE THE ANIMAL TO LOWER ALTITUDE, plus diuretics and supportive care. Locoweed worsens it.

1.4 Cardiac

Topic	Detail
Most common arrhythmia	ATRIAL FIBRILLATION. Sign: pulsus alternans — two quick normal pulses followed by a dropped pulse. Usually resolves when the underlying GI disease is corrected.
Five causes of cardiomyopathy	Gossypol · lymphosarcoma · monensin · selenium (deficiency or toxicity) · vitamin E deficiency.
Vegetative endocarditis	Organism: <i>Trueperella pyogenes</i> . The valve most commonly affected in cattle is the TRICUSPID (right side) — opposite the aortic/mitral pattern of small animals. Recommended treatment: EUTHANASIA — the prognosis is hopeless once vegetations form.
Blood types	The two clinically relevant systems in cattle are B and J . Treat a transfusion reaction with dexamethasone, epinephrine, an antihistamine, and oxygen.

RAPID-FIRE FORESTOMACH & METABOLIC RECALL

Bradycardia + indigestion → **vagal indigestion**.

Epistaxis + hemoptysis + right heart failure → **caudal vena cava thrombosis**.

High rumen chloride + "internal vomiting" → **abomasal impaction** → hypochloremic alkalosis.

Bleeding abomasal ulcer → **transfuse**.

Lactating cow on lush spring pasture, staggers → **hypomagnesemia**.

Faded coat with spectacles + diarrhea → **copper deficiency from molybdenum excess**.

Post-parturient hemoglobinuria → **phosphorus deficiency**.

PAP score >46 → **susceptible to brisket disease**.

Tricuspid vegetative endocarditis → *Trueperella* → **euthanize**.

2 | Enteric Disease & the Calf Diarrhea Algorithm

THE SINGLE MOST TESTABLE TABLE IN BOVINE MEDICINE

Calf diarrhea is answered by **AGE**. Memorize the age windows and you can answer almost any calf scours question without knowing anything else about the case.

2.1 Calf Diarrhea by Age

Agent	Age	Features
<i>E. coli</i> (ETEC)	< 4 days	The earliest scours. Secretory, watery diarrhea; profound dehydration and acidosis. K99/F5 fimbrial adhesin.
Rotavirus	5 days – 4 weeks	Villous atrophy → malabsorptive diarrhea.
Coronavirus	4 – 30 days	Damages both small intestine and colon — generally more severe than rotavirus.
Enterotoxemia (<i>C. perfringens</i> type C)	~2 weeks	Affects the HEALTHIEST, fastest-growing calf. Hemorrhagic, peracute, often found dead. That "best calf in the pen" detail is the exam hook.
<i>Cryptosporidium</i> Z	1 – 3 weeks	Chronic emaciation with TENESMUS. Zoonotic and resistant to standard disinfectants.
<i>Salmonella</i> and coccidiosis	> 21 days	The "older calf" causes. <i>Salmonella</i> brings fever and blood/mucus; coccidiosis brings tenesmus and hematochezia.

THE NON-CONTAGIOUS ONE

Clostridium perfringens causes calf diarrhea but is **NOT** transmitted between calves — it comes from soil and the normal gut flora, proliferating when management changes. The same point holds in adult cattle: *C. perfringens* is **not** contagious.

2.2 Adult Enteric Disease

Disease	Agent	Features	Diagnosis & Control
Johne's disease (paratuberculosis)	<i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i>	Two signs: chronic PROFUSE DIARRHEA in a bright, still-eating adult cow, and progressive WEIGHT LOSS. Bottle jaw from hypoproteinemia. Granulomatous enteritis with corrugated ("brain-like") intestinal mucosa and enlarged mesenteric nodes. Infection occurs in calfhood; disease appears at 2–5 years.	Fecal PCR/culture, serum ELISA. Control by preventing calfhood exposure — calve in a clean area, remove the calf immediately, feed pasteurized colostrum. Test and cull; no treatment.
Winter dysentery	CORONAVIRUS	Explosive, dark, bloody diarrhea in adult DAIRY cows , with a sharp milk drop. Occurs in JANUARY — the seasonality is the hook. Two presentations: diarrhea in neonates and this adult winter form.	Self-limiting over days; supportive care.
Salmonellosis Z	<i>Salmonella enterica</i> — most commonly S. Dublin and S. Typhimurium	Sudden, foul-smelling diarrhea often with mucus and blood , high fever, dehydration, rapid weight loss. <i>S. Dublin</i> also causes septicemia and pneumonia in calves.	Culture. Zoonotic — a real risk to handlers. Supportive care; antibiotic use is controversial.

Disease	Agent	Features	Diagnosis & Control
BVDV / mucosal disease	Pestivirus	Three presentations: acute BVD (immunosuppression, diarrhea), persistent infection (PI), and mucosal disease (a PI animal superinfected with cytopathic biotype — always fatal). An in-utero calf becomes PI when infected at 80–150 days of gestation — before immune competence. Also causes hydranencephaly and cerebellar hypoplasia.	Diagnose by PCR (ear notch antigen for PI). Identify and cull PI animals — they are the reservoir.
Coccidiosis	<i>Eimeria</i> — the most common identified cause of coccidial diarrhea in calves	Tenesmus, hematochezia in calves >3 weeks. Nervous coccidiosis causes central blindness and seizures.	Treat with amprolium plus sulfonamides. Note amprolium is a thiamine analog — high doses can precipitate PEM .
Pseudodiphtheritic membranes in the intestine	—	Think BVD (or salmonellosis).	PCR.

3 | Respiratory Disease & BRDC

3.1 The Bovine Respiratory Disease Complex

Agent	Role
The five primary VIRAL pathogens of BRDC	1. IBR (BHV-1) · 2. BVDV · 3. BRSV (bovine respiratory syncytial virus) · 4. PI-3 (parainfluenza-3) · 5. Bovine coronavirus These damage the airway and suppress immunity, opening the door for the bacteria below.
<i>Mannheimia haemolytica</i>	The principal bacterial killer in shipping fever. CRANIOVENTRAL distribution with a HEPATIZED (consolidated, liver-like) lung — the classic necropsy description. Fibrinous pleuropneumonia.
<i>Histophilus somni</i>	Causes heart failure (myocarditis with papillary muscle necrosis) as part of BRDC, plus TEME — see §4.
<i>Mycoplasma bovis</i>	Three classic signs together: respiratory signs · SWOLLEN JOINTS · HEAD TILT with suppurative EAR involvement (otitis). That triad is essentially diagnostic. Also a contagious mastitis pathogen. Notoriously refractory to treatment.
<i>Pasteurella multocida</i> and <i>Trueperella pyogenes</i>	Secondary invaders; <i>Trueperella</i> forms abscesses in chronic cases.

3.2 IBR and Other Respiratory Disease

Disease	Key Facts
IBR (infectious bovine rhinotracheitis)	Bovine herpesvirus-1 (BHV-1). Affects young cattle . Four manifestations — all the same virus: 1. Respiratory: profuse purulent nasal discharge with a RED NOSE/MUZZLE and high fever 2. Abortion in the last trimester 3. Multifocal necrotic foci in the liver and lung of the aborted fetus 4. Infectious pustular vulvovaginitis (IPV) — pustules on the vulva Note: BHV-5 is the encephalitic strain — that is how herpes 1 and herpes 5 differ in cattle.
Verminous pneumonia (lungworm)	<i>Dictyocaulus viviparus</i> — "husk." Affects young, naive cattle and yearlings , typically late summer and autumn on pasture. Life cycle: ingested larvae migrate through the gut wall → lymphatics → bloodstream → lungs → break into alveoli → crawl up the airways to the trachea → coughed up and swallowed → mature in the intestine → eggs passed in feces. Two signs: a loud husky COUGH and EOSINOPHILIA.
Fog fever (acute bovine pulmonary emphysema and edema)	The stem: a FAT COW moved to LUSH AUTUMN PASTURE, sudden respiratory distress, AFEBRILE, SQ emphysema, lungs fail to collapse on necropsy. Sick within 2–4 weeks of the pasture move. Pathophysiology: lush pasture is high in L-tryptophan → converted in the rumen to indole acetic acid → then to 3-METHYLINDOLE (3-MI) by <i>Lactobacillus</i> → absorbed → activated by lung cytochrome P450 → toxic to type I pneumocytes and Clara (club) cells → interstitial pneumonia and emphysema. Three signs: SQ emphysema · loud cough · open-mouth breathing. Brassicas (rape, kale, turnips) do the same thing by the same mechanism.
IBR vs. fog fever	IBR is FEBRILE and PURULENT; fog fever is AFEBRILE with no discharge. Of the two, IBR is the purulent one.

4 | Neurology

THE DOWN OR BLIND RUMINANT — WORK THIS LIST

PEM · lead · listeriosis · TEME · rabies · nervous coccidiosis · hypomagnesemia · hypovitaminosis A · polioencephalomalacia from sulfur.

The top two differentials for CNS signs and blindness in cows are PEM and LEAD.

Disease	Features	Diagnosis & Treatment
Polioencephalomalacia (PEM)	Cortical blindness (central — menace absent but PLR intact), head pressing, star-gazing, seizures. Bilaterally symmetric. Three causes: 1. THIAMINE (vitamin B₁) deficiency — thiaminase-producing rumen flora 2. SULFUR toxicity (high-sulfate water, distillers grains) — the two most common causes in cows are thiamine deficiency and excess sulfur 3. High-dose AMPROLIUM — a thiamine analog In horses the plant cause is BRACKEN FERN (thiaminase).	Thiamine IV — response is rapid and is itself diagnostic. Necropsy: fluorescence of cortex under UV light, laminar cortical necrosis.
Lead poisoning	Four signs: acute BLINDNESS · dullness · down and unable to rise · unaware of surroundings. Batteries, old paint, crankcase oil. Calves are curious and most affected.	Calcium EDTA (or succimer). Do NOT use sodium EDTA — it binds calcium. A public-health issue: milk and meat residues.
Listeriosis ("circling disease")	<i>Listeria monocytogenes</i> from spoiled silage (pH >5.5). Ascends the trigeminal nerve to the brainstem. UNILATERAL cranial nerve deficits — dropped ear, dropped lip, drooling, dysphagia, and a decrease in facial sensation. Circling toward the lesion.	CSF: MONONUCLEAR pleocytosis with elevated total protein. Treat with PROCAINE PENICILLIN, high dose and prolonged. Also causes late-term abortion. Zoonotic.
TEME (thromboembolic meningoencephalitis)	<i>Histophilus somni</i> (formerly <i>Haemophilus somnus</i>) causing septicemia and vasculitis. FEVER and RESPIRATORY signs come FIRST, before the CNS signs — that sequence is the discriminator from listeriosis. Brainstem and cerebellar signs. Feedlot cattle.	CSF: NEUTROPHILIC pleocytosis with XANTHOCHROMIC (yellow) fluid — indicating prior hemorrhage. Oxytetracycline or florfenicol.
Rabies R Z	Abnormal BELLOWING is the classic bovine sign, plus tenesmus, drooling, ataxia, and aggression or paralysis.	Always on the differential. Report; extreme caution with oral exams.
Meningitis in calves	<i>E. coli</i> — almost always secondary to failure of passive transfer and septicemia.	Antibiotics that cross the blood-brain barrier; plasma.
Nervous coccidiosis	Central blindness, which can persist for months, with seizures. A neurotoxin-mediated syndrome accompanying <i>Eimeria</i> infection.	Treat the coccidiosis; supportive care.
Hydranencephaly — three viruses	BVD · Bluetongue · Border disease. (Add Akabane and Cache Valley.) In-utero infection destroys developing cerebrum.	Prevention by vaccination and vector control.
Arthrogryposis	The two VIRAL causes: BLUETONGUE and AKABANE. The plant cause is LUPINE (crooked calf disease) at 40–70 days of gestation.	Prevention.

LISTERIOSIS VS. TEME — THE CSF ANSWERS IT

LISTERIA: unilateral cranial nerve deficits, **MONONUCLEAR** CSF, silage history → **procaine penicillin**.

TEME: fever and respiratory disease **first**, then brainstem/cerebellar signs, **NEUTROPHILIC and XANTHOCHROMIC** CSF, feedlot → oxytetracycline.

RAPID-FIRE RUMINANT NEUROLOGY RECALL

Blind cow, head pressing → **PEM** → **thiamine** (causes: thiaminase, sulfur, amprolium).

Blind, dull, down, unaware → **lead** → **calcium EDTA**.

Dropped lip, circling, silage → **Listeria** → mononuclear CSF → **procaine penicillin**.

Fever and pneumonia first, then brainstem signs → **TEME** → neutrophilic xanthochromic CSF.

Abnormal bellowing → **rabies**.

Calf with meningitis → **E. coli** after FPT.

Hydranencephaly → **BVD, bluetongue, Border disease**.

5 | Reproduction, Abortion & the BSE

5.1 Pregnancy Diagnosis Timeline

Method	When	Notes
Palpation of the amniotic vesicle / "membrane slip"	30–35 days	The earliest reliable rectal finding. Rough handling of the vesicle at this stage is associated with ATRESIA COLI in the calf.
Placentomes palpable	70 days	Cotyledon-caruncle units.
Fremitus	120–150 days	Vibration ("thrill") of the middle uterine artery as blood flow increases.
Gestation	280 days / 9 months	Estrus 18 hours; cycle 21 days; year-round polyestrus.

5.2 Abortion — Organized by Timing

"BIG COWS ABORT LATE" — THE LATE-TERM ABORTION MNEMONIC

Brucella · *Lepto* · *Listeria* — the classic **last-trimester** abortifacients. Add **IBR** and **ponderosa pine needles**.

If time is short, the "three Bs" of bovine abortion: **BRUCELLA** · **BVD** · *Bacillus/bacterial* (and *campylobacter*).

Cause	Timing	Distinguishing Features
<i>Brucella abortus</i> R Z	Last trimester	The classic late abortion with a RETAINED PLACENTA . ZOONOTIC (undulant fever). Screening: the MILK RING TEST on bulk tank milk. Vaccinate heifers at 4–12 months with RB51. Orange metal tags are placed in the RIGHT ear of vaccinated heifers.
<i>Leptospira</i> (hardjo) Z	Late	Late abortion + WEAK CALVES + AGALACTIA. Also causes " flabby bag " mastitis — a sudden drop in milk with a soft, flaccid udder and no inflammation. Zoonotic.
<i>Listeria monocytogenes</i> Z	Late	Late abortion; also the neurologic disease. Silage-associated.
<i>Neospora caninum</i>	Any time; peak 4–7 months	The leading cause of bovine abortion worldwide. A protozoal parasite. The DOG is the definitive host — transmission by feed or water contaminated with dog feces, plus efficient vertical transmission in the herd.
IBR (BHV-1)	Last trimester	Aborted fetus has multifocal white NECROTIC FOCI in the LIVER and LUNG.
<i>Tritrichomonas foetus</i>	1–8 months	REPEAT BREEDER / early empty cow. Bulls are ASYMPTOMATIC, LIFELONG carriers (the organism lives in preputial crypts of older bulls). STRICT CULLING IS REQUIRED — there is no effective vaccine.
<i>Campylobacter fetus</i> (vibriosis)	Mid-gestation, 4–7 months	Also a repeat breeder . Highly effective vaccines are available and given annually — the key contrast with <i>Tritrichomonas</i> . Diagnose by PCR. Common in the western US.
Mycotic abortion	Late	Two signs: a THICK, LEATHERY PLACENTA, and round RAISED RING-LIKE PLAQUES AROUND THE EYES of the fetus. <i>Aspergillus</i> and zygomycetes. Brucella can also give a thickened placenta but does NOT give the facial plaques — the plaques are what separate them.
<i>Ureaplasma diversum</i>	Any	Causes granular vulvitis and abortion . A common normal inhabitant of the vagina, which is why interpretation requires care.

Cause	Timing	Distinguishing Features
Epizootic bovine abortion ("foothill abortion")	Late	Vector: <i>Ornithodoros coriaceus</i> — a SOFT tick. The fetus is aborted with LARGE lymph nodes and spleen , and a DESTROYED THYMUS . California foothills.
Ponderosa pine needles	Last trimester	Isocupressic acid; abortion with retained placenta.

5.3 Postpartum & Periparturient Disease

Condition	Key Facts
Retained fetal membranes	Caused by dystocia, twins, abortion, hypocalcemia, selenium/vitamin E deficiency, and induced parturition . The placenta is normally expelled within 8–12 hours in the cow ; beyond 24 hours it is retained.
Metritis	Three predisposing factors: DYSTOCIA · KETOSIS · HYPOCALCEMIA . Treat with systemic antibiotics and prostaglandin F2-alpha.
Pyelonephritis	<i>Corynebacterium renale</i> . Occurs in the MULTIPAROUS cow around 2.5 months post-partum . Ascending infection; hematuria, pyuria, colic. Treat with PENICILLIN (long course).
Vaginal vs. uterine prolapse	VAGINAL prolapse occurs BEFORE calving (late gestation, heritable, recurs). UTERINE prolapse occurs IMMEDIATELY AFTER calving — an emergency associated with hypocalcemia.
Persistent hymen	Segmental aplasia of the paramesonephric (Müllerian) ducts — same in horses.
Estrus behavior after standing heat	An open heifer will show estrus again starting around day 18–24 from an observed standing heat (a 21-day cycle).

5.4 The Breeding Soundness Examination (BSE)

Element	Detail
The two cultures/PCRs on a bull BSE	<i>Tritrichomonas foetus</i> and <i>Campylobacter fetus</i> — preputial scrapings. Both are venereal, both cause repeat breeding, and both live silently in the bull.
Scrotal circumference	Correlates with daily sperm output — the single best predictor of a bull's fertility, and the reason it is measured on every BSE.

5.5 Production Targets

Parameter	Target
Calving interval	12–13 months (365–395 days).
First service conception rate	~50–60% in dairy cattle.
Average days open	85–115 days .
Services per conception	1.5–2.0 .
Interpretation	A LOW first-service conception rate together with a HIGH services-per-conception points to a problem with heat detection, insemination technique, or semen handling — not with the cows' fertility per se. Poor breeding technique specifically decreases the first-service conception rate .

6 | Mastitis & the Udder

Category	Detail
CONTAGIOUS mastitis pathogens	<i>Mycoplasma bovis</i> and <i>Streptococcus agalactiae</i> — spread cow-to-cow at milking. Add <i>Staphylococcus aureus</i> . Control: milking hygiene, post-dipping, dry cow therapy, and culling. <i>Mycoplasma</i> does not respond to antibiotics — cull.
ENVIRONMENTAL mastitis pathogens	Coliforms and environmental streptococci — from bedding and manure. Control by environment and bedding management.
Severe clinical coliform mastitis	The three MacConkey-positive organisms: <i>E. coli</i>, <i>Klebsiella</i>, and <i>Enterobacter</i>. These produce endotoxic shock, a watery secretion, and a cold quarter.
"Flabby bag" mastitis	LEPTOSPIROSIS — sudden agalactia with a soft, flaccid udder and no heat or pain, often with blood-tinged milk and fever.
Teat-end hyperkeratosis	Most common cause: over-milking / excessive machine vacuum and liner issues . A milking-machine management problem.
Milker's nodules 	Pseudocowpox (a parapoxvirus) — nodules on the teats that are infectious to people , producing "milker's nodules" on the hands.
Udder impetigo / ulcerative dermatitis	Foul-smelling ulcers on the udder of post-partum dairy cattle — necrotic dermatitis, often <i>Fusobacterium/Trueperella</i> . Hygiene and topical/systemic therapy.
Contagious agalactia (small ruminants)	<i>Mycoplasma agalactiae</i> . Three signs: KCS · arthritis · mastitis. Management: CULL.

MILK RESIDUES — THE REGULATORY QUESTIONS

SULFONAMIDES are prohibited in lactating dairy cattle (with the narrow exception of sulfadimethoxine, which is labeled).

Chloramphenicol is completely banned in all food animals. Extralabel use of **fluoroquinolones** is prohibited in food animals, and **cephalosporins** are restricted. Always check withdrawal times — a residue violation is a career problem as much as a clinical one.

Why we pasteurize milk: to kill *Mycobacterium bovis*, *Brucella*, *Listeria*, *Salmonella*, *Campylobacter*, *E. coli* O157, and *Coxiella burnetii* — the raw milk zoonoses.

7 | Musculoskeletal, Foot & Lameness

Condition	Agent	Features & Treatment
Foot rot (interdigital phlegmon)	<i>Fusobacterium necrophorum</i>	Symmetric interdigital swelling with a foul-smelling necrotic fissure; sudden severe lameness and fever. Responds well to systemic antibiotics. Same organism as calf diphtheria and liver abscesses .
Interdigital dermatitis	<i>Fusobacterium necrophorum</i>	Superficial inflammation of the interdigital skin; a precursor to heel horn erosion.
Digital dermatitis ("hairy heel warts", Mortellaro)	<i>Treponema</i> — a spirochete	The discriminator: digital dermatitis is a painful, red, strawberry-like erosion at the SKIN-HORN JUNCTION of the HEEL, usually on the plantar aspect of the hind foot , often with hair-like keratin projections. Interdigital dermatitis is between the claws. Control: footbaths (copper sulfate/formalin) and topical tetracycline.
Calf diphtheria	<i>Fusobacterium necrophorum</i>	Calves < 3 months. GRAY-TO-YELLOW DIPHTHERITIC MEMBRANES in the mouth/pharynx, foul breath, fever, dysphagia. Treat with PENICILLIN. Can extend to the larynx and obstruct.
Lumpy jaw	<i>Actinomyces bovis</i> — gram POSITIVE	BONE. A hard, immovable swelling of the MANDIBLE or maxilla , with draining tracts containing "sulfur granules." Osteomyelitis — poor response to treatment. Iodides ± antibiotics.
Wooden tongue	<i>Actinobacillus lignieresii</i>	SOFT TISSUE. A hard, painful, protruding TONGUE with difficulty prehending. Responds well to sodium iodide IV.
Gastrocnemius rupture	—	Described as " HOCK DOWN but HIPS OK " — the hock drops to the ground while the animal can still bear weight cranially. Usually associated with calving or slipping.
White muscle disease	Selenium/vitamin E deficiency	Affects skeletal and CARDIAC muscle ; elevated CK and AST . Also in horses, pigs, small ruminants, and poultry.

LUMPY JAW VS. WOODEN TONGUE

Lumpy jaw = *ActinoMYCES* = **bone** ("MY jaw BONE") → poor prognosis.

Wooden tongue = *ActinoBACILLUS* = **soft tissue** → responds to **sodium iodide**.

Both are gram-positive-appearing filamentous organisms with sulfur granules; the tissue involved is the discriminator.

8 | Dermatology

Condition	Agent	Features & Treatment
Ringworm Z	<i>Trichophyton verrucosum</i>	The most common dermatophyte of cattle. Thick gray-white crusts, classically around the EYES and FACE in CALVES. Seen in WINTER when cattle are housed. NOT itchy. Treatment: none needed — it is self-limiting over 1–4 months. Highly zoonotic to handlers.
Rain rot / dermatophilosis Z	<i>Dermatophilus congolensis</i> — a gram-positive facultative anaerobe	Crusted "paintbrush" lesions after prolonged wetting. Cytology: "RAILROAD TRACKS" — parallel rows of cocci. ZINC DEFICIENCY is a contributing factor. Keep dry.
Lumpy skin disease FAD	Capripoxvirus	Firm skin nodules; a foreign animal disease.
Pediculosis (lice)	Sucking: <i>Haematopinus</i> Biting/chewing: <i>Bovicola bovis</i>	Pruritus with hair loss and dirty debris and dirt in the coat . Winter problem. Pour-on macrocyclic lactones or pyrethroids.
Hypoderma (cattle grubs / warbles)	<i>Hypoderma bovis</i> , <i>H. lineatum</i>	Treat with POUR-ON MOXIDECTIN. Critical timing rule: do NOT treat while larvae are migrating through the esophagus or spinal canal — killing them there causes bloat or paralysis. Treat after the fall heel-fly season and before migration.
Sarcoptic mange Z	<i>Sarcoptes scabiei</i>	Treat with POUR-ON DORAMECTIN ; dairy cows with moxidectin pour-on . Burrowing mite — needs systemic therapy.
<i>Psoroptes</i> and <i>Chorioptes</i>	Surface mites	Both live ON the skin, so topical sprays and dusts work. <i>Psoroptes</i> — common in cattle, sheep, and rabbits; reportable in sheep . <i>Chorioptes</i> — leg mange in cattle and horses; long legs, short pedicles .
Stephanofilariasis	<i>Stephanofilaria</i>	Plaque-like dermatitis along the VENTRAL MIDLINE. Transmitted by the HORN FLY (<i>Haematobia irritans</i>) .
Photosensitization	—	Primary: the plant itself is photodynamic — St. John's wort, Hypericum, Erodium, brassicas, and tetracyclines. Secondary (hepatogenous): liver damage prevents clearance of PHYLLERYTHRIN , the chlorophyll breakdown product, which accumulates in skin and reacts with UV light. Affects white/unpigmented areas.

9 | Vector-Borne, Systemic & Reportable Disease

9.1 Vector-Borne Disease

THE FIVE VECTOR-BORNE DISEASES OF CATTLE

1. *Anaplasma marginale* 2. *Babesia* 3. Heartwater 4. Epizootic bovine abortion 5. *Otobius megnini*

Disease	Agent	Vector	Features & Treatment
Anaplasmosis	<i>Anaplasma marginale</i>	<i>Dermacentor</i>	Found INSIDE the RBC (at the margin). Causes EXTRAVASCULAR hemolysis — which is why you see ICTERUS and PALE mucous membranes but NO hemoglobinuria / NO red urine . Also Howell-Jolly bodies (from the regenerative response). Treat with OXYTETRACYCLINE.
Babesiosis ("tick fever", "Texas fever", "redwater")	<i>Babesia bovis</i> and <i>B. bigemina</i>	<i>Rhipicephalus</i> (Boophilus)	PEAR-SHAPED organisms inside RBCs. Causes INTRAVASCULAR hemolysis → severe anemia, HEMOGLOBINURIA (red water), fever, and jaundice. The stem: "redwater hemoglobinuria in Texas cows near Mexico" → BABESIA . Eradicated from the US by tick control; imidocarb where permitted.
Theileriosis	<i>Theileria</i> spp. (<i>T. orientalis</i>)	Asian longhorned tick — <i>Haemaphysalis longicornis</i>	Severe hemolytic anemia , fever, jaundice, and reduced milk yield . An emerging US problem with the spread of the longhorned tick.
Heartwater FAD	<i>Ehrlichia ruminantium</i>	<i>Amblyomma</i> (ornate, long mouthparts)	FATAL ENCEPHALITIS in sheep, goats, and cattle. Necropsy: PERICARDIAL EFFUSION — the "hydropericardium" that gives the disease its name. A foreign animal disease.
Epizootic bovine abortion ("foothill abortion")	A deltaproteobacterium	<i>Ornithodoros coriaceus</i> — a SOFT tick	Late-term abortion; fetus has enlarged lymph nodes and spleen with a DESTROYED THYMUS .
<i>Otobius megnini</i> (spinose ear tick)	—	A SOFT tick	Predilection site: the EARS . Causes otitis and irritation.
Trypanosomiasis ("Nagana", African animal trypanosomiasis)	<i>Trypanosoma congolense</i> — the most common and important cause of AAT in East and Central Africa	TSETSE FLY (<i>Glossina</i>) — cyclical biological transmission	Three signs: intermittent FEVER · severe ANEMIA · WEIGHT LOSS.
Bluetongue R	Orbivirus	<i>CULICOIDES</i> — the biting midge	In SHEEP (most severe): oral lesions/erosions, vasculitis, edema of the muzzle, tongue, and CORONARY BANDS , and arthrogryposis with hydranencephaly in utero. Cattle are usually subclinical reservoirs.
Bovine leukosis	BLV — a retrovirus	Biting flies , needles, rectal sleeves, colostrum	Persistent lymphocytosis; lymphosarcoma in a minority. Also a cause of cardiomyopathy.
Louping ill	FLAVIVIRUS	<i>Ixodes</i>	Ovine encephalomyelitis . Two signs: fine muscle tremors and "nervous nibbling." UK/Ireland.

ANAPLASMA VS. BABESIA VS. LEPTOSPIROSIS VS. BACILLARY HEMOGLOBINURIA VS. ANTHRAX

ANAPLASMA — EXTRAvascular hemolysis → **icterus and pallor, but NO red urine.**

BABESIA — INTRAvascular → **RED URINE (hemoglobinuria).**

Bacillary hemoglobinuria (*C. haemolyticum*) — **red urine + a liver infarct**; follows *Fasciola* migration.

LEPTOSPIROSIS — **hemoglobinuria plus abortion and agalactia.**

ANTHRAX — **sudden death with unclotted blood and NO rigor mortis.**

9.2 Systemic & Reportable Disease

Disease	Agent	Features, Diagnosis & Action
Anthrax R Z	<i>Bacillus anthracis</i>	Sudden death. Necropsy findings: LACK OF RIGOR MORTIS and UNCLOTED blood oozing from orifices. DIAGNOSIS: obtain VITREOUS HUMOR or a BLOOD SMEAR FROM AN EAR SCRAPE — do NOT open the carcass. Opening it exposes the vegetative organism to oxygen and creates environmentally persistent spores. Three forms in humans: intestinal · pulmonary · cutaneous. Reportable and zoonotic.
Bovine tuberculosis R Z	<i>Mycobacterium bovis</i>	Chronic weight loss, cough, enlarged lymph nodes, CASEOUS CALCIFIED GRANULOMAS, and "PEARL DISEASE" on the pleura. Screening: the CAUDAL FOLD TEST — intradermal bovine tuberculin in the caudal tail fold, read at 72 hours. To differentiate <i>M. bovis</i> from <i>M. avium</i>: the COMPARATIVE CERVICAL TEST. Definitive: CULTURE (slow — takes weeks). ZOONOTIC, REPORTABLE, test-and-slaughter, NO treatment.
Brucellosis R Z	<i>Brucella abortus</i>	Late-term abortion with retained placenta. Milk ring test for herd screening; RB51 vaccination at 4–12 months; orange metal tag in the RIGHT ear. Zoonotic (undulant fever) — a classic occupational infection of veterinarians via vaccine needle-stick.
Malignant catarrhal fever (MCF) R	Ovine herpesvirus-2 (sheep-associated) or alcelaphine herpesvirus-1 (wildebeest)	Six signs: high fever, bilateral corneal opacity/uveitis, profuse oculonasal discharge, erosive stomatitis, generalized lymphadenopathy, and diarrhea. The classic history is cattle housed near SHEEP. Essentially always fatal.
Foot and Mouth Disease FAD R	Aphthovirus	Vesicles on the tongue, muzzle, teats, and coronary bands in cloven-hoofed animals. Horses are NOT affected — that is the key discriminator from vesicular stomatitis.
Vesicular stomatitis R Z	Vesiculovirus	Affects three species: PIGS, HORSES, and SHEEP (and cattle). The vesicular disease that DOES affect horses — which is how you distinguish it from FMD in the field. Zoonotic.
Rinderpest FAD	Morbillivirus	The "four Ds": DEPRESSION · DISCHARGES (ocular and nasal) · DIARRHEA · DEATH. Severe fever, erosive mouth lesions, profuse diarrhea, extreme dehydration. Globally eradicated in 2011 — but still examinable. Differentiate from Rift Valley fever: RVF is a mosquito-borne bunyavirus causing abortion storms and hepatic necrosis, and is zoonotic; rinderpest is a directly transmitted morbillivirus with erosions and diarrhea.
BSE R Z	Prion	"Mad cow disease." Progressive neurologic disease in adult cattle. Feed bans; ruminant-to-ruminant protein prohibition.
Scrapie R	Prion	Sheep and goats. Test by IMMUNOHISTOCHEMISTRY of the OBEX (also third eyelid and rectal mucosa biopsy in live animals). National eradication program with mandatory identification.

Disease	Agent	Features, Diagnosis & Action
Peste des petits ruminants (PPR) FAD	Reportable MORBILLIVIRUS (paramyxovirus) — related to rinderpest and canine distemper	Four signs: fever · oral and nasal lesions · oculonasal discharge · stomatitis and conjunctivitis. Action: REPORT IT. Diagnose by PCR.

THE FIVE CLASSIC REPORTABLE CATTLE DISEASES

MCF · Bluetongue · Vesicular stomatitis · Rinderpest · Foot and Mouth Disease.

Add **anthrax, bovine TB, brucellosis, BSE, and scrapie** — all reportable, all test-and-slaughter or eradication programs, none treated.

10 | Parasitology

10.1 Gastrointestinal Nematodes

Parasite	Location	Disease
<i>Haemonchus contortus</i> ("barber pole worm")	ABOMASUM	A blood-sucking strongyle. Causes localized bleeding → severe ANEMIA and SUBMANDIBULAR EDEMA ("bottle jaw") from hypoproteinemia — with NORMAL fecal consistency. That last detail is the classic trap: haemonchosis does NOT cause diarrhea. Primarily sheep and goats. Monitor with FAMACHA scoring of conjunctival color.
<i>Ostertagia ostertagi</i> ("brown stomach worm")	ABOMASUM	Causes abomasal pH to RISE above 6 (destroying parietal glands), giving diarrhea, weight loss, and hypoproteinemia. Necropsy: "COBBLESTONE" or "MOROCCAN LEATHER" appearance of the abomasal mucosa. Type I: disease from a current adult infection (grazing season). Type II: disease from the synchronous emergence of ARRESTED (hypobiotic) larvae — late winter/spring, in older animals, and much harder to treat.
<i>Trichostrongylus</i>	Abomasum/SI	The third of the "three stomach worms of cattle" with <i>Haemonchus</i> and <i>Ostertagia</i> .
<i>Oesophagostomum</i>	Large intestine / colon	"Nodular worm" — forms nodules in the intestinal wall.
<i>Nematodirus</i>	Small intestine	Diarrhea, weight loss, failure to thrive — especially in young lambs and crias. Diagnose by direct smear and fecal flotation.
<i>Dictyocaulus viviparus</i>	Lungs (bronchi)	Verminous pneumonia / "husk" in yearlings. Identify with the BAERMANN technique — you are looking for larvae, not eggs. Treat with ivermectin.
<i>Muellerius capillaris</i>	Small bronchioles (sheep/goats)	Causes verminous pneumonia with small firm gray-green NODULES on the lung surface. <i>Dictyocaulus filaria</i> is the MOST PATHOGENIC lungworm of small ruminants ; <i>Muellerius</i> is the most common but less pathogenic.
<i>Setaria cervi</i>	Peritoneal cavity	Usually an incidental finding at slaughter; aberrant migration can cause neurologic disease.
<i>Thelazia gulosa</i>	EYE	Eyeworm — conjunctivitis and keratitis; face fly transmitted.

10.2 Trematodes & Cestodes

Parasite	Detail
<i>Fasciola hepatica</i> (liver fluke)	A TREMATODE (flatworm). Intermediate host: LYMNAEID SNAILS. Acute: migratory tract hemorrhage and hepatitis. Chronic (adults in bile ducts): weight loss, ANEMIA, hypoproteinemia, SUBMANDIBULAR EDEMA (bottle jaw), and poor wool/production. Two classic sequelae: 1. "BLACK DISEASE" — infectious necrotic hepatitis from <i>Clostridium novyi</i> type B. Fluke migration creates the anaerobic niche in which clostridial spores germinate. 2. BACILLARY HEMOGLOBINURIA / red water disease from <i>C. haemolyticum</i> — the same mechanism.
<i>Eurytrema</i> (pancreatic fluke)	Parasitizes the pancreatic and biliary ducts of ruminants. Light infections are asymptomatic; heavy burdens cause bovine eurytrematosis.

Parasite	Detail
<i>Paramphistomum</i> (rumen fluke)	Adults in the rumen are largely harmless; immature flukes migrating through the duodenum cause severe enteritis in heavy infections.
<i>Moneizia</i>	The tapeworm of ruminants. Largely non-pathogenic but alarming to owners when proglottids appear. Treat with fenbendazole. Oribatid mite intermediate host.
<i>Echinococcus granulosus</i> Z	A cestode whose definitive host is the DOG . Sheep and other ruminants are intermediate hosts and develop HYDATID CYSTS in the LIVER and LUNGS . Fatal hydatid disease in humans. Do not feed raw offal to farm dogs.

10.3 Protozoa & Other

Parasite	Detail
<i>Eimeria</i>	4 sporocysts. Infects ruminants, horses, rabbits, and poultry — NOT dogs and cats . The most common identified cause of coccidial diarrhea in calves. Treat with amprolium plus sulfonamides. In sheep and goats: monensin as a coccidiostat in feed, and amprolium in water for the sick.
<i>Cryptosporidium</i> Z	Acid-fast or IFA staining required; resistant to disinfectants and chlorination. Calves 1–3 weeks; chronic emaciation with tenesmus. Zoonotic — a genuine risk to students and farm children.
<i>Neospora caninum</i>	Leading cause of abortion worldwide; dog is the definitive host ; efficient vertical transmission.
<i>Toxoplasma gondii</i>	Major abortifacient in sheep and goats ; does NOT cause abortion in COWS . Cat is the definitive host. Placental lesions: bright dark red cotyledons with white foci of necrosis.
<i>Tritrichomonas foetus</i>	Affects the REPRODUCTIVE system. Bulls are lifelong carriers; culling is the only control. PCR of preputial scrapings is the best test.
<i>Parelaphostrongylus tenuis</i> ("meningeal worm")	Definitive host: the WHITE-TAILED DEER. Aberrant migration in sheep, goats, and camelids causes ataxia, paraparesis, torticollis, and head tilt. CSF: EOSINOPHILIC pleocytosis with elevated protein — the diagnostic hallmark.
<i>Oestrus ovis</i>	Nasal bot of sheep — larvae in the nasal passages and sinuses cause sneezing and nasal discharge ("false gid").
<i>Onchocerca</i>	Carried by <i>Culicoides</i> .

RAPID-FIRE RUMINANT PARASITOLOGY RECALL

Bottle jaw + anemia + **normal manure** → *Haemonchus* (abomasum) → FAMACHA.
 Cobblestone abomasum + diarrhea + pH >6 → *Ostertagia*; type II = arrested larvae emerging.
 Bottle jaw + snail + liver → *Fasciola* → sequelae **black disease** and **bacillary hemoglobinuria**.
 Husky cough in a yearling → *Dictyocaulus* → **Baermann** → ivermectin.
 Gray-green lung nodules in a sheep → *Muellerius*.
 Ataxic goat or alpaca with eosinophilic CSF → **meningeal worm** (white-tailed deer host).
 Abortion in sheep but not cows → *Toxoplasma*.
 Repeat breeder with a carrier bull → *Tritrichomonas* → **cull**.

11 | Small Ruminant Medicine

11.1 The Lentiviruses — CAE and OPP

ONE VIRUS FAMILY, TWO SPECIES, NEARLY IDENTICAL DISEASE

CAE (goats) and OPP / Maedi-visna (sheep) are both LENTIVIRUSES — the same family as FIV and HIV. Both are **transmitted in COLOSTRUM and milk**, cause **lifelong infection**, and are controlled only by **test and cull plus feeding heat-treated colostrum**. Neither has a treatment or a vaccine.

Disease	Species	Clinical Syndromes	Diagnosis
CAE (caprine arthritis encephalitis)	GOATS	Three forms by age: 1. Neurologic signs in 2–4 month olds (leukoencephalomyelitis) 2. ARTHRITIS in animals >1 year — swollen carpi ("big knee") 3. HARD UDDER (indurative mastitis) with agalactia Also chronic pneumonia and wasting.	ELISA or AGID for screening. Transmitted in COLOSTRUM — infected for life.
OPP (ovine progressive pneumonia) / Maedi-visna	SHEEP	Four signs: PNEUMONIA (progressive dyspnea, "maedi" = shortness of breath), ATAXIA ("visna" = wasting/neurologic), WEIGHT LOSS , and HARD UDDER . The two cardinal signs are respiratory disease and INDURATIVE MASTITIS .	PCR — test and cull. The virus is Maedi virus , another lentivirus.

11.2 Clostridial Diseases of Sheep and Goats

Organism / Type	Disease	Treatment & Prevention
<i>C. perfringens</i> type A	Alpha toxin only. Causes hemorrhagic bowel syndrome in adult cattle and YELLOW LAMB DISEASE .	General treatment: PENICILLIN · ANTITOXIN · IV FLUIDS · NSAIDs. Prevention: pre-farrow/pre-lambing vaccination of dams so that antibody is transferred in colostrum. CD-T vaccine covers types C and D plus tetanus.
<i>C. perfringens</i> type B	LAMB DYSENTERY — but NOT present in the US.	
<i>C. perfringens</i> type C	Produces alpha PLUS BETA toxin . Causes PERACUTE DEATH with hemorrhagic enteritis in 1–3 day old kids and lambs . Why neonates specifically: they do not yet produce the pancreatic proteolytic enzyme TRYPSIN , which would otherwise destroy beta toxin. Colostrum also contains trypsin inhibitors. Called " struck " in Great Britain.	
<i>C. perfringens</i> type D	ENTEROTOXEMIA — "PULPY KIDNEY" disease . Epsilon toxin. Classic in the fastest-growing lamb on high grain; sudden death, opisthotonos, glycosuria.	
<i>C. novyi</i> type B	"BIG HEAD" — toxin-mediated necrosis causing edema and swelling of the head in young rams that fight. Also "BLACK DISEASE" / infectious necrotic hepatitis , which follows Fasciola migration creating anaerobic hepatic niches. Peracute, sudden-death presentation.	PENICILLIN . Control flukes to prevent black disease.
<i>C. haemolyticum</i>	Bacillary hemoglobinuria / "red water" — also follows fluke migration.	Penicillin; fluke control.
<i>C. tetani</i>	Tetanus — spastic paralysis.	PENICILLIN , antitoxin, wound care.

11.3 Small Ruminant Infectious Disease

Disease	Agent	Features & Management
Contagious ecthyma ("ORF", sore mouth) Z	PARAPOXVIRUS	Crusty scabs and cauliflower-like proliferative growths around the MOUTH, NOSTRILS, EYELIDS, and HOOVES (and on the ewe's teats, causing her to reject the lamb). ZOONOTIC — causes painful nodules on handlers' hands. Self-limiting in 1–4 weeks; live scab vaccine available but commits the flock.
Caseous lymphadenitis (CLA)	<i>Corynebacterium pseudotuberculosis</i>	Abscessation of peripheral (and internal) lymph nodes with characteristic onion-ring, laminated caseous pus . Diagnosis: CULTURE . Lifelong; cull or isolate. Zoonotic risk to handlers.
Chlamydial polyarthritis	<i>Chlamydia pecorum</i>	Three signs: LAMENESS · CONJUNCTIVITIS · FEVER . Treat with OXYTETRACYCLINE .
Enzootic abortion of ewes (EAE)	<i>Chlamydia abortus</i> Z	Late-term abortion. Placental lesions: necrotic COTYLEDONS and INTERCOTYLEDONARY SPACE . Seriously zoonotic — can cause abortion in pregnant women . Pregnant women must not handle lambing ewes .
Mycoplasma polyarthritis	<i>Mycoplasma</i> spp.	Three outcomes: death in the young · brain disease · polyarthritis .
Vibriosis	<i>Campylobacter fetus</i> subsp. <i>fetus</i>	A cause of abortion, common in the WESTERN USA . Reduce by VACCINATION .
Toxoplasmosis	<i>Toxoplasma gondii</i>	Abortion with bright dark-red cotyledons bearing white foci of necrosis — a distinctive placental picture.
Brucellosis R	<i>Brucella ovis</i>	In rams: EPIDIDYMITIS and decreased fertility . Palpate every ram's epididymis at breeding soundness exam.
Ulcerative posthitis ("pizzle rot")	<i>Corynebacterium renale</i>	Urea-splitting bacteria on a high-protein diet produce ammonia that ulcerates the prepuce. Reduce dietary protein.
Lumpy wool	<i>Dermatophilus congolensis</i>	The sheep form of dermatophilosis — crusts binding the wool into hard tufts.
Sheep keds	<i>Melophagus ovinus</i>	A wingless fly, not a tick. Anemia and wool damage.
Interdigital dermatitis / foot rot	<i>Fusobacterium necrophorum</i> (with <i>Dichelobacter nodosus</i>)	Footbaths (zinc sulfate), trimming, and culling chronically affected animals.
<i>Psoroptes ovis</i> R	Surface mite	The most important sheep mite and REPORTABLE — eradicated from the US . Found on the back, sides, and rump . Non-burrowing , with an oval body and long pedicles .
Border disease ("hairy shaker disease")	PESTIVIRUS (related to BVDV)	Lambs born with tremors and a hairy birth coat , plus hydranencephaly. Diagnose by virus isolation .
Louping ill	Flavivirus; <i>Ixodes</i>	Ovine encephalomyelitis ; fine muscle tremors and nervous nibbling.
Systemic pasteurellosis	<i>Mannheimia/Pasteurella</i>	Treat with oxytetracycline or florfenicol .

11.4 Small Ruminant Metabolic & Nutritional Disease

Condition	Detail
Pregnancy toxemia	Late-gestation ewe or doe carrying multiple fetuses in negative energy balance → ketosis. Treatment: deliver the fetuses — C-SECTION or induction — because the fetal glucose demand is the problem. Plus propylene glycol, dextrose, and calcium.

Condition	Detail
Copper toxicosis Z —the sheep-specific poisoning	SHEEP are uniquely sensitive to copper; PIGS are resistant. The classic cause: feeding CATTLE or HORSE feed to sheep — those rations contain copper levels that are safe for cattle and lethal for sheep. Copper accumulates silently in the liver, then releases in a hemolytic crisis. Signs: HEMOLYTIC ANEMIA with methemoglobinemia, hemoglobinemia, and hemoglobinuria. Necropsy: GUNMETAL-colored kidneys and "PORT WINE" urine. Treat with D-PENICILLAMINE and AMMONIUM TETRATHIOMOLYBDATE plus fluids to increase excretion.
Copper DEFICIENCY	Two syndromes, separated by age: ENZOOTIC ATAXIA — lambs 1–2 months old, acquired postnatally. SWAYBACK — congenital or in very young lambs, with progressive ascending paralysis. Both are demyelinating. Also caused by molybdenum excess.
Zinc deficiency in goats	Parakeratosis and poor coat quality.
PEM	Causes in small ruminants: HIGH SULFUR and LOW THIAMINE; also high-dose AMPROLIUM. Treat with thiamine.
Frothy bloat	Sign: OPEN-MOUTH BREATHING with severe left-sided distension. Legume pastures. Treat with poloxalene or mineral oil; trocarize if life-threatening.
Polled intersex	Three hornless goat breeds are affected: Saanen, Alpine, and Toggenburg. The polled gene is linked to intersex — breeding polled to polled produces sterile intersex offspring.

TWO SMALL RUMINANT DRUG DANGERS

XYLAZINE causes FATAL HYPOXEMIA in sheep and goats — small ruminants are exquisitely sensitive to alpha-2 agonists, which cause pulmonary edema and severe hypoxemia. Use with great caution and at much reduced doses.

AMPROLIUM at high doses causes PEM in sheep — it is a thiamine analog. Dose carefully and supplement thiamine on prolonged courses.

RAPID-FIRE SMALL RUMINANT RECALL

Big knee in a goat + hard udder → **CAE** → colostrum-transmitted lentivirus.

Progressive pneumonia + hard udder in a sheep → **OPP/Maedi-visna**.

Sudden death in the best-grown lamb on grain → **C. perfringens type D pulpy kidney**.

Peracute hemorrhagic death in a 1–3 day old lamb → **type C** (no trypsin yet).

Swollen head in a fighting ram → **C. novyi type B big head** → penicillin.

Scabby mouth, zoonotic → **ORF (parapoxvirus)**.

Onion-ring abscess → **caseous lymphadenitis** → culture.

Gunmetal kidneys + port wine urine → **copper toxicosis** (fed cattle feed).

Ewe with twins, off feed, ketotic → **pregnancy toxemia** → **C-section**.

Hairy shaker lamb → **Border disease**.

12 | Camelids

CAMELIDS ARE PSEUDO-RUMINANTS — THE ANATOMY IS DIFFERENT

Camelids have a **three-compartment stomach (C1, C2, C3)**, not four. **C1 gets IMPACTIONS and gastroliths; C3 gets ULCERS.** That single distinction answers several questions.

12.1 Anatomy, Physiology & Reproduction

Topic	Detail
Stomach compartments	C1 — impactions and GASTROLITHS. C3 — ULCERS (and the site of acid secretion). Remember: " C1 blocks, C3 burns. "
Most common congenital malformation	CHOANAL ATRESIA — the cria cannot breathe through the nose (and camelids are obligate nasal breathers), so it presents with respiratory distress and inability to nurse.
Ovulation	INDUCED ovulators — with cats, rabbits, and ferrets.
Pregnancy detection	As early as 16 days .
Uterine horn	Essentially ALL pregnancies occur in the LEFT uterine horn — which is why UTERINE TORSION is such a common camelid dystocia.
Placentation	EPITHELIOCHORIAL, MICROCOTYLEDONARY, DIFFUSE. Fetal villi are spread over the entire chorionic sac (diffuse) but attach through thousands of tiny villous clusters (microcotyledonary) . Because it is epitheliochorial it is NON-INVASIVE — all maternal and fetal tissue layers remain intact, giving a relatively loose attachment. No antibody crosses in utero — crias depend entirely on colostrum.
Puberty	Llamas reach puberty later than alpacas — roughly 2–3 years for llamas versus 12–18 months for alpacas (females).

12.2 Camelid Medicine

Condition	Detail
Three causes of anemia	1. C3 ULCER (GI blood loss) 2. <i>Haemonchus contortus</i> 3. <i>Mycoplasma haemolamae</i> (formerly <i>Eperythrozoon</i>)
<i>Mycoplasma haemolamae</i>	A hemotropic bacterium that attaches to the SURFACE of red blood cells. Many infected animals are asymptomatic carriers, but the organism triggers severe, sometimes fatal hemolytic anemia when the animal is stressed, immunosuppressed, or has a concurrent illness. Treat with OXYTETRACYCLINE.
Ulcers vs. impactions	C3 ulcers — reduce by minimizing stress, providing adequate roughage, and using prophylactic anti-ulcer medication in hospitalized animals. C1 impactions — from poor-quality forage and dehydration.
Most common parasites	NEMATODES — strongyles, whipworms, and the meningeal worm . Also <i>Eimeria</i> coccidiosis (including <i>E. macusaniensis</i> , which is highly pathogenic). Treat Eimeria with PONAURIL.
<i>Nematodirus</i>	Three signs: diarrhea · weight loss · failure to thrive. Diagnose by direct smear and fecal float.
Meningeal worm	<i>Parelaphostrongylus tenuis</i> — causes meningoencephalitis , ataxia, and paresis. White-tailed deer is the definitive host. Prevent by keeping deer off pasture and monthly ivermectin in endemic areas.
Enterotoxemia	<i>C. perfringens</i> types A, C, and D. Vaccinate.
Alpaca fever	<i>Streptococcus equi</i> subsp. <i>zooepidemicus</i> — septicemia with high fever, often after stress or shows.
Juvenile llama immunodeficiency syndrome	Very low IgG from a genetic defect in B lymphocyte production / humoral immunity. Recurrent infections and failure to thrive despite adequate colostrum.

Condition	Detail
Berserk male syndrome	Also called NOVICE HANDLER SYNDROME — a hand-reared male that has imprinted on humans becomes dangerously aggressive at maturity. Prevention is everything: do not bottle-raise and over-handle male crias.
Ear ticks	Cause otitis media and interna .

12.3 Camelid Dermatology

Condition	Detail
Munge	Idiopathic hyperkeratosis — thick crusted lesions around the nose, mouth, eyes, and ears . Affects animals 6 months to 2 years old. Multifactorial; rule out zinc deficiency and mites first.
Zinc-responsive dermatosis	Non-pruritic papules and crusting; responds to zinc supplementation.
Sarcoptic mange	<i>Sarcoptes scabiei</i> var. <i>cameli</i> — a BURROWING mite. Treat with IVERMECTIN . Remember: <i>Psoroptes</i> and <i>Chorioptes</i> live ON the skin and can be treated topically.
<i>Psoroptes ovis</i> R	REPORTABLE in camelids . Causes lesions in the EARS . Camelids serve as ASYMPTOMATIC CARRIERS — which is exactly why it is reportable and why they matter epidemiologically to the sheep industry.

TILMICOSIN IS CARDIOTOXIC IN CAMELIDS

Tilmicosin (Micotil) is fatal in camelids — and in horses, goats, swine, and humans. It causes dose-dependent cardiovascular collapse with no effective antidote. **Never use it in a camelid**, and handle the bottle carefully — accidental human self-injection has been fatal.

RAPID-FIRE CAMELID RECALL

Impaction → **C1**. Ulcer → **C3**.

Cria that can't breathe through its nose → **choanal atresia**.

All pregnancies in the **left horn** → uterine torsion risk.

Placenta: **epitheliochorial, microcotyledonary, diffuse**.

Anemia → **C3 ulcer** · *Haemonchus* · *Mycoplasma haemolamae* → oxytetracycline.

Ataxic alpaca → **meningeal worm** → eosinophilic CSF.

Reportable ear mite, asymptomatic carrier → *Psoroptes ovis*.

Aggressive hand-raised male → **berserk male syndrome**.

Never give tilmicosin.

13 | Toxicology

13.1 Reproductive & Teratogenic Toxins

THE FIVE REPRODUCTIVE TOXINS

1. False hellebore (cyclopamine) 2. Locoweeds 3. Fescue 4. Lupine 5. Ponderosa pine

Plant	Toxin	Effect	Critical Timing
False hellebore <i>Veratrum californicum</i> (and skunk cabbage)	CYCLOPAMINE — a teratogen	Four defects: CYCLOPIA · anophthalmos · cleft palate · metacarpal hypoplasia. The classic "monkey-faced lamb."	DAY 14 of gestation → craniofacial deformities (cyclopia). DAY 30 → limb and bone shortening of the metacarpal/metatarsal region.
Lupine	Anagryrine	"CROOKED CALF DISEASE" — ARTHROGRYPOSIS (joint stiffness/contracture) and CLEFT PALATE , plus neurologic disease.	Days 40–70 of gestation.
Ponderosa pine needles	Isocupressic acid	ABORTION in the LAST TRIMESTER , with retained placenta and weak calves.	Third trimester. Also "western yellow pine" and "white pine."
Locoweeds <i>Astragalus</i> , <i>Oxytropis</i> , <i>Swainsona</i>	SWAINSONINE	Inhibits alpha-mannosidase , the lysosomal enzyme that recycles spent glycoproteins → the cell cannot clear them → NEUROVISCERAL CYTOPLASMIC VACUOLATION → neurologic disease and ABORTION . Also worsens high-altitude/brisket disease .	Found in the West and Rocky Mountains.
Fescue	ERGOPEPTINE ALKALOIDS — DOPAMINE AGONISTS	PROLACTIN SUPPRESSION → inhibits lactation. Prolonged gestation and reproductive failure in mares. Four general signs: lameness · sloughing of the rear hooves · decreased weight gain · fat necrosis. "Summer slump" and "fescue foot" in cattle.	Treat with DOMPERIDONE — a dopamine ANTAGONIST , which restores prolactin.
Zearalenone	An ESTROGENIC mycotoxin (<i>Fusarium</i>)	PIGS are the sensitive species — hyperestrogenism and pseudopregnancy. See the swine guide.	Moldy corn.

13.2 Hepatic Toxins

Toxin	Detail
Pyrrolizidine alkaloids	Four plants: fiddleneck (<i>Amsinckia intermedia</i>) · ragwort · groundsel · <i>Senecio</i>. Mechanism: hepatocytes cannot divide → progressive liver failure and hepatic encephalopathy. Three diagnostic findings: MEGALOCYTOSIS · PERIORTAL FIBROSIS · BILE DUCT HYPERPLASIA. Often causes SECONDARY photosensitization. Prognosis: POOR — the damage is cumulative and irreversible by the time signs appear.
Aflatoxin	A mycotoxin from <i>Aspergillus</i>, found in MOLDY GRAIN (especially <i>A. flavus</i> in corn). Hepatotoxic, immunosuppressive, and carcinogenic.

Toxin	Detail
Poison hemlock <i>Conium maculatum</i>	Toxin: CONIINE — a nicotinic agonist causing a depolarizing block. PATHOGNOMONIC SIGN: the cow's BREATH and URINE smell of MOUSE URINE ("mousey odor"). Also a teratogen causing arthrogryposis.

13.3 Pulmonary Toxins

Toxin	Detail
The three pulmonary toxins: PERILLA MINT · MOLDY SWEET POTATOES (4-ipomeanol) · BRASSICAS (3-methylindole). All three produce atypical interstitial pneumonia with wet, firm lungs that fail to collapse at necropsy.	
Perilla mint	Damages two cell types: TYPE I PNEUMOCYTES and BRONCHIOLAR (Clara/club) EPITHELIAL CELLS . Necropsy: WET, EMPHYSEMATOUS lungs. HIGH MORTALITY due to pulmonary fibrosis .
Moldy sweet potatoes	Produce 4-IPOMEANOL → atypical interstitial pneumonia . Postmortem: lungs are wet, firm, and fail to collapse — identical to the 3-MI picture.
Brassicas (rape, kale, turnips)	High L-TRYPTOPHAN → 3-METHYLINDOLE → FOG FEVER . See §3.2.

13.4 Cardiac, Hematologic & Metabolic Toxins

Toxin	Detail
Four cardiac toxin plants	Milkweed · gossypol (cotton seed) · OLEANDER (a cardiac glycoside) · foxglove. Arrhythmias and sudden death. Gossypol toxicity is inactivated by IRON.
Ionophores	Monensin, lasalocid, salinomycin. Used in cattle as a COCCIDIOSTAT and to improve feed efficiency. HORSES are the toxic species — myocardial necrosis and dilated cardiomyopathy. Keep cattle feed away from horses.
Six coagulopathy toxins	Bracken fern · cyanide · carbon monoxide · nitrates · moldy sweet clover · anthrax.
Bracken fern	Two toxins: PTAQUILOSIDE (a toxic glycoside) and THIAMINASE . In CATTLE: severe BONE MARROW SUPPRESSION — low neutrophils and lymphocytes and depleted platelets → acute hemorrhagic disease ; plus chronic BOVINE ENZOOTIC HEMATURIA (bladder tumors and bleeding). In HORSES: POLIOENCEPHALOMALACIA from thiaminase. One plant, two completely different diseases in two species.
Moldy sweet clover	COUMARIN is converted to DICOUMAROL — a vitamin K antagonist (warfarin-like) → spontaneous bleeding and hemorrhage. Vitamin K is required for factors II, VII, IX, and X. Diagnose with the PT , because factor VII (extrinsic pathway) has the shortest half-life and falls first.
Cyanide	Five plants: chokecherry · arrow grass · Sudan grass · Johnson grass · sorghum. Blocks cellular respiration and oxidative transport — hemoglobin cannot release oxygen. Three signs: BRIGHT RED mucous membranes · BRIGHT RED blood · BITTER ALMOND smell of the GI tract on necropsy. Three treatments: SODIUM THIOSULFATE · SODIUM NITRITE · METHYLENE BLUE.
Nitrate	BROWN blood and methemoglobinemia — the direct contrast with cyanide's bright red. Treat with METHYLENE BLUE. Nitrate accumulates in drought-stressed forage and fertilized pasture.
Carbon monoxide	The other cause of cherry-red mucous membranes .
Blue-green algae	Three names: <i>Anabaena</i> , <i>Microcystis</i> , <i>Aphanizomenon</i> . Birds and cattle are the most commonly affected. A cow can die within MINUTES. Two pathways: nicotinic agonist (mimics ACh) and acetylcholinesterase inhibitor; toxin absorbed through mucous membranes causes a post-synaptic depolarizing block . Neurotoxic signs: miosis, ptialism, bradycardia, diarrhea, ataxia, convulsions, death.

Toxin	Detail
Salt poisoning / water deprivation	Lesion: PERIVASCULAR INFILTRATION OF EOSINOPHILS — pathognomonic, and best known in pigs. Pathogenesis: high blood sodium raises osmolarity → water leaves the brain → the brain generates idiogenic osmoles to draw water back → when the animal is suddenly given free water, the brain over-swells → head pressing, star-gazing, blindness, seizures. Also causes HEMOLYSIS as water rushes into red cells. Treatment principle: reintroduce water SLOWLY.
Bovine bonkers (NPN / ammoniated forage toxicity)	Non-protein nitrogen toxicity — urea or ammoniated feed. The toxic compound is 4-methylimidazole. Hyperexcitability, tremors, stampeding. Mechanism: urea is hydrolyzed to AMMONIA, which RAISES rumen pH (alkaline) — and the alkaline pH promotes further ammonia absorption, driving the toxicity. Treatment: ruminal infusion of 2–8 L of ACETIC ACID (vinegar) plus 3–10 L of COLD WATER — acidifying the rumen traps ammonia as non-absorbable ammonium, and the cold water slows fermentation.
Lead	Four signs: acute blindness · dullness · down and unable to rise · unaware of surroundings. From batteries, paint, and crankcase oil. Chelate with CALCIUM EDTA or succimer — never SODIUM EDTA.
Oak / acorns	Tannins. Five signs: hemorrhagic diarrhea · abdominal pain · tenesmus · colic · death. In CATTLE the main lesion is RENAL damage. In calves it causes severe nephrosis. Sheep and especially GOATS are the LEAST sensitive — they tolerate tannins better.
Water hemlock	RESINOIDS (cicutoxin) — the LEAVES and roots are toxic. In cattle: salivation, twitching, dilated pupils, and death. Five outcomes: grand mal seizures · salivation · tachypnea · birth defects · death. One of the most acutely lethal plants in North America.
Nightshade	Atropine-like — parasympatholytic. Treat with physostigmine (which itself causes dilated pupils, progressive paralysis, and depression in excess).
Rhododendron / azalea	GRAYANOTOXIN — causes GI signs, cardiac arrhythmias, and it CANNOT be metabolized. Vomiting (in species that can), bradycardia, hypotension.
Cocklebur	PIGS are the most affected species; the seedling (two-leaf) stage is toxic. Causes acute hepatic necrosis with hypoglycemia.
Slaframine	MOLDY RED CLOVER. Causes hypersalivation — "slobbers."
Ergotism	<i>Claviceps purpurea</i>, a parasitic fungus on rye, oat, wheat, and Kentucky bluegrass — and, unusually, on stored grain. Pathophysiology: alkaloid → vascular constriction → thrombosis → GANGRENE of the extremities, ears, and tail, plus colic. Distinguish from FESCUE toxicosis, which is caused by the endophyte <i>Epichloë coenophiala</i> in tall fescue and produces the same vasoconstrictive "fescue foot" — the mechanism is shared but the fungus is different.
Larkspur (<i>Delphinium</i>)	Nerve block at the neuromuscular junction, muscle failure, SEVERE BLOAT — very lethal and not fixable. A leading cause of cattle death on western rangeland.
Yellow star thistle	Nigropallidal encephalomalacia — a HORSE disease, not a ruminant one. Keep the species attached.

14 | Production Medicine, Anatomy & Regulatory

14.1 Microbiology Mnemonics

Group	Members
Gram-positive organisms <i>"Bacteria Like Chocolate Cookies"</i>	B acillus · L isteria · C lostridium and C orynebacterium · C occi (staph, strep, enterococcus)
Gram-positive BACILLI <i>"MACDONALD"</i>	M ycobacterium · A ctinomyces · C lostridium · C orynebacterium · N ocardia · A nthraxis · L isteria · D iphtheroids
Gram stain colors	Gram POSITIVE = PURPLE. Gram NEGATIVE = PINK.

14.2 Urolithiasis & Miscellaneous

Topic	Detail
Most common urolith on an all-grain diet	STRUVITE (magnesium ammonium phosphate) — from high phosphorus and alkaline urine. The classic feedlot steer or wether.
Most common urolith in the western US	SILICATE — from silica-rich range grasses.
Nutritional note	During high milk production, the two amino acids rumen microbes most need are LYSINE and METHIONINE — the standard rumen-protected supplements.
Measure of nitrogen metabolism	MUN — milk urea nitrogen — reflects the protein-to-energy balance of the ration.
Poikilocytosis	NORMAL in GOATS (and young cattle) — do not over-interpret it.

Appendix A | Reportable & Foreign Animal Diseases

THE ANSWER IS ALMOST ALWAYS "CALL THE AUTHORITIES"

Notify the **State Animal Health Official** or the **USDA Area Veterinarian in Charge**, restrict movement, and do not treat. If a question offers a treatment alongside "contact the state veterinarian," choose the latter.

Disease	Agent	Recognition	Action
Anthrax R Z	<i>Bacillus anthracis</i>	Sudden death, no rigor mortis, unclotted blood.	Do NOT open the carcass — vitreous humor or ear-scrape smear.
Bovine TB R Z	<i>M. bovis</i>	Caseous calcified granulomas, pearl disease.	Caudal fold test at 72 h; test and slaughter.
Brucellosis R Z	<i>B. abortus</i> , <i>B. ovis</i>	Late abortion, retained placenta; ram epididymitis.	Milk ring test; RB51; orange right-ear tag.
Foot and Mouth Disease FAD	Aphthovirus	Vesicles; cloven-hoofed only — NOT horses .	Immediate report; depopulate.
Vesicular stomatitis R Z	Vesiculovirus	Pigs, horses, sheep — the vesicular disease that DOES affect horses.	Report.
Rinderpest FAD	Morbillivirus	The four Ds: depression, discharges, diarrhea, death.	Globally eradicated; still examinable.
Peste des petits ruminants FAD	Morbillivirus	Fever, oral/nasal lesions, oculonasal discharge, stomatitis.	Report; diagnose by PCR.
Malignant catarrhal fever R	OvHV-2	Corneal opacity, erosions, lymphadenopathy; sheep contact.	Report; fatal.
Bluetongue R	Orbivirus	Coronary band and oral lesions in sheep; <i>Culicoides</i> .	Report; vector control.
Heartwater FAD	<i>Ehrlichia ruminantium</i>	Fatal encephalitis; pericardial effusion; <i>Amblyomma</i> .	Report.
BSE R Z	Prion	Progressive neurologic disease in adults.	Report; feed ban compliance.
Scrapie R	Prion	Pruritus, wasting, ataxia in sheep/goats.	IHC of the OBEX ; national eradication program.
Psoroptes ovis R	Surface mite	Sheep scab; camelids are asymptomatic carriers.	Report; eradicated from the US.
Screwworm R	<i>Cochliomyia hominivorax</i>	Larvae in living tissue.	Report immediately.
Lumpy skin disease FAD	Capripoxvirus	Firm cutaneous nodules.	Report.
Rabies R Z	Lyssavirus	Abnormal bellowing, tenesmus, paralysis.	Report; public health.

Appendix B | Numbers, Ages & Timelines

Number	Meaning
Cow gestation 280 days (9 months)	Estrus 18 hours; cycle 21 days; year-round polyestrus.
Sheep/goat gestation 150 days	Estrus 24 hours; cycle ~17 days; seasonally polyestrus (short-day).
Palpation slip 30–35 d · placentomes 70 d · fremitus 120–150 d	Bovine pregnancy timeline.
PI window: 80–150 days of gestation	When BVDV in-utero infection creates a persistently infected calf.
Lupine: days 40–70	Crooked calf disease.
Cyclopamine: day 14 (face), day 30 (limbs)	False hellebore teratogenesis.
Calving interval 12–13 months	Days open 85–115; services per conception 1.5–2.0; first-service conception ~50–60%.
Rumen pH 6	Falls to 5 in grain overload.
Caudal fold TB test read at 72 hours	Comparative cervical test differentiates <i>M. bovis</i> from <i>M. avium</i> .
RB51 at 4–12 months	Brucella vaccination; orange tag in the right ear.
PAP score <41 mmHg good, >46 susceptible	High-altitude/brisket disease screening.
Calf scours by age	<i>E. coli</i> <4 d · rota 5 d–4 wk · corona 4–30 d · <i>C. perfringens</i> ~2 wk · crypto 1–3 wk · salmonella and coccidia >21 d.
Ruminant teeth: 32	No upper incisors — a dental pad instead.
Camelid pregnancy detectable at 16 days	All pregnancies in the left horn.
Paravertebral block: T13, L1, L2	Needles over the transverse processes of L1, L2, L4.